

# Computer Assignment 3

Ziyi Wang ( [ID removed] )

Min Zhu ( [ID removed] )

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## Exercise 3:1

### 3:1.1

We start with the saturated model and remove interaction terms in a systematic way. To satisfy the requirement of removing parameters in order, we go from saturated model msat to the all three-way interactions model m3 and then from all three-way interactions model m3 to all two-way interactions model m2. And to get all models, we do the forward selection from simplest model m1 to the all two-way interactions model m2. The goodness-of-fit of the different models is evaluated with deviance (compared to the saturated model) and AIC, and Table 4 shows the calculated values.

Table 1: Model selection table

| Model             | Deviance | df | p-value    | AIC    |
|-------------------|----------|----|------------|--------|
| XYZV              | 0        | 0  | 1          | 123.97 |
| XYZ,XYV,XZV,YZV   | 0.35935  | 1  | 0.5488675  | 122.33 |
| XYV,XZV,YZV       | 0.41133  | 2  | 0.8141058  | 120.39 |
| XYV,XZV           | 0.63838  | 3  | 0.8875926  | 118.61 |
| XYV               | 1.16271  | 4  | 0.8841992  | 117.14 |
| XY,XZ,XV,YZ,YV,ZV | 1.7225   | 5  | 0.8860530  | 115.70 |
| XY,XZ,XV,YV,ZV    | 1.8221   | 6  | 0.9353113  | 113.80 |
| XY,XV,YV,ZV       | 4.7486   | 7  | 0.6906113  | 114.72 |
| XY,XV,ZV          | 7.7197   | 8  | 0.4613186  | 115.69 |
| XY,ZV             | 17.845   | 9  | 0.0370150  | 123.82 |
| ZV                | 35.45    | 10 | 0.0001046  | 139.43 |
| X,Y,Z,V           | 377.79   | 11 | 3.1518e-74 | 479.76 |

### 3:1.2

From Table 4, we select the model containing XY, XZ, XV, YV, and ZV as the final model. This model achieves the lowest AIC value of 113.80 among all candidate models, indicating an optimal balance between goodness-of-fit and model complexity. Furthermore, it yields the highest p-value of 0.9353113, which is well above the 0.05 threshold, suggesting that the model fits the data well and cannot be rejected. The deviance of 1.8221 is relatively small given the 6 degrees of freedom. Additionally, the model includes only five two-way interaction terms and avoids higher-order interactions, making it both parsimonious and straightforward to interpret. Hence, the model XY, XZ, XV, YV, ZV is selected.

### 3:1.3

For the log-linear model we selected, we have that:

$$\log(\mu_{ijkl}) = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_l^V + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{il}^{XV} + \lambda_{jl}^{YV} + \lambda_{kl}^{ZV}, i, j, k, l \in \{0, 1\}.$$

where X = Mother's age, Y = Smoking habits, Z = Gestational age, and V = Child survival.  $\mu_{ijkl}$  = expected value of number given the children's condition X=i, Y=j, Z=k, V=l.

#### Parameter Estimates and Significance Analysis:

Based on the model output (Table 2), the parameter estimates, standard errors, z-values, and p-values are as follows:

Table 2: Parameter estimates for the selected model (XY, XZ, XV, YV, ZV)

| Parameter   | Estimate | Std. Error | z value | Pr           |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 3.94312  | 0.12624    | 31.236  | $< 2e - 16$  |
| X           | -0.29199 | 0.17033    | -1.714  | 0.08648      |
| Y           | -1.71313 | 0.24310    | -7.047  | $1.83e - 12$ |
| Z           | -0.77237 | 0.18237    | -4.235  | $2.28e - 05$ |
| V           | 1.81422  | 0.13442    | 13.497  | $< 2e - 16$  |
| Z:V         | 3.31135  | 0.18452    | 17.945  | $< 2e - 16$  |
| X:Y         | -0.41132 | 0.09950    | -4.134  | $3.56e - 05$ |
| X:V         | -0.46481 | 0.18003    | -2.582  | 0.00983      |
| Y:V         | -0.44375 | 0.24471    | -1.813  | 0.06977      |
| X:Z         | -0.16557 | 0.09599    | -1.725  | 0.08456      |

#### Odds Ratio Analysis:

Based on the parameter estimates from the log-linear model, we calculate odds ratios (OR) and their 95% confidence intervals to quantify the strength of associations between variables (Table 3):

Table 3: Odds ratios and 95% confidence intervals for the selected model

| Parameter                               | OR     | 95% CI           |
|-----------------------------------------|--------|------------------|
| X (Mother's age)                        | 0.747  | (0.535, 1.043)   |
| Y (Smoking habits)                      | 0.180  | (0.112, 0.290)   |
| Z (Gestational age)                     | 0.462  | (0.323, 0.660)   |
| V (Child survival)                      | 6.136  | (4.715, 7.986)   |
| X:Y (Age $\times$ Smoking)              | 0.663  | (0.545, 0.806)   |
| X:Z (Age $\times$ Gestational age)      | 0.847  | (0.702, 1.023)   |
| X:V (Age $\times$ Survival)             | 0.628  | (0.441, 0.894)   |
| Y:V (Smoking $\times$ Survival)         | 0.642  | (0.397, 1.037)   |
| Z:V (Gestational age $\times$ Survival) | 27.422 | (19.100, 39.370) |

#### Model Interpretation:

##### 1. Main Effects

1.1 Child survival (V): Child survival status is the most significant factor (OR = 6.136, 95% CI: 4.715-7.986,  $p < 0.001$ ), indicating that surviving children are far more common than deceased children in the data.

1.2 Smoking habits (Y): Mothers smoking 5+ cigarettes per day have significantly lower expected counts compared to those smoking less than 5 cigarettes per day (OR = 0.180, 95% CI: 0.112-0.290,  $p < 0.001$ ).

1.3 Gestational age (Z): Gestational age of 260+ days is associated with significantly lower counts compared to less than 260 days (OR = 0.462, 95% CI: 0.323-0.660,  $p < 0.001$ ).

## 2. Interaction Effects

### 2.1 Significant Interactions

#### 2.1.1 Gestational age and child survival interaction (Z:V):

This is the strongest interaction effect (OR = 27.422, 95% CI: 19.100-39.370,  $p < 0.001$ ).

Gestational age has a strong protective effect on child survival: infants with gestational age  $\geq 260$  days have 27.4 times higher survival odds compared to those with gestational age  $< 260$  days. And the confidence interval of (19.100, 39.370) does not include 1, confirming the association is statistically significant.

#### 2.1.2 Mother's age and smoking habits interaction (X:Y):

Significant negative interaction (OR = 0.663, 95% CI: 0.545-0.806,  $p < 0.001$ ).

The association between mother's age and smoking habits is not independent; specifically, older mothers (30+ years) are less likely (0.663 times) to be in the low-smoking category ( $< 5$  cigarettes/day) compared to what would be expected under independence, after controlling for other variables. And the confidence interval is entirely below 1, indicating a consistent negative association.

#### 2.1.3 Mother's age and child survival interaction (X:V):

Significant negative interaction (OR = 0.628, 95% CI: 0.441-0.894,  $p = 0.0098$ ).

Indicates that older mothers (30+ years) have 37.2% lower infant survival advantage compared to younger mothers.

Note that the main effect of mother's age is not significant, so this effect is primarily manifested through the interaction.

### 2.2 Non-Significant Interactions

#### 2.2.1 Smoking habits and child survival interaction (Y:V):

Non-Significant negative interaction (OR = 0.628, 95% CI: 0.441-0.894,  $p = 0.0698$ ).

Although the point estimate suggests reduced survival odds for infants of mothers smoking 5+ cigarettes/day, the p-value exceeds 0.05, indicating insufficient evidence for this association in the presence of other variables.

#### 2.2.2 Mother's age and gestational age interaction (X:Z):

Not significant (OR = 0.847, 95% CI: 0.702-1.023,  $p = 0.0846$ ).

The confidence interval includes 1, suggesting no clear evidence of an association between maternal age and gestational age in this model.

## 3:1.4

In this question, we set  $V = \text{Child survival}$  as the response variable, and  $X = \text{Mother's age}$ ,  $Y = \text{Smoking habits}$ ,  $Z = \text{Gestational age}$  as explanatory variables to build up a logistic regression model.

To determine whether the logistic regression model should include two-way or three-way interactions among  $X$ ,  $Y$ ,  $Z$ , we perform goodness-of-fit assessments using the stepwise approach.

The three model structures considered are as follows:

1. Main Effects Only Model (ml1):

This model includes only the main effects of the predictors  $\mathbf{x}$ ,  $\mathbf{y}$ , and  $\mathbf{z}$ .

$$\text{logit}(P(\mathbf{v} = 1|\mathbf{x}, \mathbf{y}, \mathbf{z})) = \beta_0 + \beta_x \mathbf{x} + \beta_y \mathbf{y} + \beta_z \mathbf{z}$$

2. Second-Order Interaction Model (ml2):

This model includes all second-order interaction terms among  $\mathbf{x}$ ,  $\mathbf{y}$ , and  $\mathbf{z}$ .

$$\text{logit}(P(\mathbf{v} = 1|\mathbf{x}, \mathbf{y}, \mathbf{z})) = \beta_0 + \beta_x \mathbf{x} + \beta_y \mathbf{y} + \beta_z \mathbf{z} + \beta_{xy} \mathbf{xy} + \beta_{xz} \mathbf{xz} + \beta_{yz} \mathbf{yz}$$

3. Saturated Model (ml3):

This model includes the highest-order interaction term (third-order) possible for these three predictors.

$$\text{logit}(P(\mathbf{v} = 1|\mathbf{x}, \mathbf{y}, \mathbf{z})) = \beta_0 + \beta_x \mathbf{x} + \beta_y \mathbf{y} + \beta_z \mathbf{z} + \beta_{xy} \mathbf{xy} + \beta_{xz} \mathbf{xz} + \beta_{yz} \mathbf{yz} + \beta_{xyz} \mathbf{xyz}$$

The results of the stepwise selection can be presented in the table below. It shows that the model containing only the main effects(*ml1*) achieves the lowest AIC, and is therefore chosen as the final model in this selection process.

Table 4: Logistic model selection table

| Model      | Deviance | df | p_value   | AIC   |
|------------|----------|----|-----------|-------|
| XYZ        | 0        | 0  | 1         | 48.86 |
| XY, XZ, YZ | 0.35935  | 1  | 0.5488675 | 47.22 |
| XY, XZ     | 0.58575  | 2  | 0.7461154 | 45.45 |
| XZ         | 0.94802  | 3  | 0.8138267 | 43.81 |
| X, Y, Z    | 1.4875   | 4  | 0.8288534 | 42.35 |

We also take Likelihood Ratio(LR) tests and all p-values are greater than 0.05, indicating that there is no necessary need to use the complex model(*ml2* and *ml3*).

Analysis of Deviance Table

```
Model 1: cbind(success, failure) ~ x + y + z
Model 2: cbind(success, failure) ~ (x + y + z)^2
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         4      1.48753
2         1      0.35935 3    1.1282  0.7703
Analysis of Deviance Table
```

```
Model 1: cbind(success, failure) ~ (x + y + z)^2
Model 2: cbind(success, failure) ~ (x + y + z)^3
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         1      0.35935
2         0      0.00000 1    0.35935  0.5489
```

Figure 1: Result of LR test

Some key statistics about this chosen model are:

Table 5: Statistics of ml1

| Coefficients | Estimate | OR( $e^{\text{Estimate}}$ ) | Std. Error | 95%CI           | p_value     |
|--------------|----------|-----------------------------|------------|-----------------|-------------|
| Intercept    | 1.8139   | 6.1343                      | 0.1351     | (1.555, 2.086)  | $< 2e - 16$ |
| X            | -0.4675  | 0.6266                      | 0.1804     | (-0.819,-0.110) | 0.00954     |
| Y            | -0.4228  | 0.6552                      | 0.2624     | (-0.917,0.117)  | 0.10711     |
| Z            | 3.3098   | 27.3784                     | 0.1846     | (2.955, 3.680)  | $< 2e - 16$ |

Therefore, plugging in all estimates, our fitted model is

$$\text{logit}(P(\mathbf{v} = 1|\mathbf{x}, \mathbf{y}, \mathbf{z})) = 1.8139 - 0.4675\mathbf{x} - 0.4228\mathbf{y} + 3.3098\mathbf{z}$$

Interpretation:

1. Mother's Age(X) has a significant negative effect on child survival:  $\beta_x$  is -0.4675 and its Odds ratio is 0.6266, and it means with gestational age(Z) and smoking habits(Y) under control, for every one-unit increase in mother's age, the odds of child survival reduces by around 37%. Moreover, this is a significant negative relationship because the corresponding p-value(0.00954) is less than 0.05.
2. Smoking habits(Y) has a non-significant negative effect on child survival:  $\beta_y$  is -0.4228 and its Odds ratio is 0.6552. This negative relationship indicates that smoking reduces the odds of child survival by around 34.5% with the other variables controlled. However, the p-value(0.10711) for this coefficient is higher than 0.05, showing that the effect of smoking habits on child survival is not statistically significant.
3. Gestational age(Z) has a strong and positive influence on child survival:  $\beta_z$  is 3.3098 and its Odds ratio is 27.3784. This means that controlling for mother's age and smoking habits, every one-day growth in gestational age leads to the odds of child survival multiplied by around 27. And it demonstrates highly significance with p-value( $< 2e - 16$ ) being very small. These two characteristics makes gestational age the most crucial factor influencing child survival in our model.

### 3:1.5

The logistic regression model we built in question 4 is

$$\text{logit}(P(\mathbf{v} = 1|\mathbf{x}, \mathbf{y}, \mathbf{z})) = \log \frac{P(\mathbf{v} = 1|x, y, z)}{P(\mathbf{v} = 0|x, y, z)} = \beta_0 + \beta_x\mathbf{x} + \beta_y\mathbf{y} + \beta_z\mathbf{z}$$

and the equivalent log-linear model is supposed to include two sets of terms: the first is the saturation model of the explanatory variables(X, Y, Z); and the second component is the interaction terms between V and X, Y, Z.

Thus the log-linear model has the following structure:

$$\begin{aligned} \log(\mathbf{m}_{xyzv}) &= \underbrace{\Lambda_P}_{\mathbf{x} \times \mathbf{y} \times \mathbf{z} \text{ terms}} + \underbrace{\Lambda_V(l)}_{V \text{ interaction terms}} \\ &= \lambda^0 + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ} + \lambda_{ijk}^{XYZ} + \lambda_l^V + \lambda_{li}^{VX} + \lambda_{lj}^{VY} + \lambda_{lk}^{VZ} \end{aligned}$$

We can prove that these two models have identical estimates and standard errors of the corresponding parameters both theoretically and practically.

#### Proof in theory:

The log odds ratio in log-linear model is

$$\begin{aligned} \log \left( \frac{\mathbf{m}_{xyz1}}{\mathbf{m}_{xyz0}} \right) &= [\Lambda_P + \Lambda_V(1)] - [\Lambda_P + \Lambda_V(0)] \\ &= \Lambda_V(1) - \Lambda_V(0) \end{aligned}$$

$$= (\lambda_1^V + \lambda_{1i}^{VX} + \lambda_{1j}^{VY} + \lambda_{1k}^{VZ}) - (\lambda_0^V + \lambda_{0i}^{VX} + \lambda_{0j}^{VY} + \lambda_{0k}^{VZ})$$

The term  $\Lambda_P$  cancels out, leaving only items related to variable  $V$ . This demonstrates that the log-linear model has the same log-odds structure as the logistic regression model. And since these two models are fitting the same conditional likelihood function, their MLEs and associated Standard Errors are numerically identical.

The corresponding coefficient equivalences between the two models are shown below:

Table 6: Corresponding coefficients of two models

| LR model                    | Log-linear model          | Equation                                  |
|-----------------------------|---------------------------|-------------------------------------------|
| Intercept $\hat{\beta}_0$   | $\hat{\lambda}_1^V$       | $\hat{\beta}_0 = \hat{\lambda}_1^V$       |
| Coefficient $\hat{\beta}_x$ | $\hat{\lambda}_{1i}^{VX}$ | $\hat{\beta}_x = \hat{\lambda}_{1i}^{VX}$ |
| Coefficient $\hat{\beta}_y$ | $\hat{\lambda}_{1j}^{VY}$ | $\hat{\beta}_y = \hat{\lambda}_{1j}^{VY}$ |
| Coefficient $\hat{\beta}_z$ | $\hat{\lambda}_{1k}^{VZ}$ | $\hat{\beta}_z = \hat{\lambda}_{1k}^{VZ}$ |

Next, we fit the log-linear models in R and evaluate the results to see if they match.

### Proof in practice:

We build a log-linear model following the structure above, and its statistics are:

Table 7: Statistics of log-linear model

| Term        | Estimate | Std. Error | z value | Pr(>  z ) |     |
|-------------|----------|------------|---------|-----------|-----|
| (Intercept) | 3.93498  | 0.12738    | 30.891  | < 2e-16   | *** |
| x1          | -0.27621 | 0.17328    | -1.594  | 0.11092   |     |
| y1          | -1.65646 | 0.26139    | -6.337  | 2.34e-10  | *** |
| z1          | -0.76156 | 0.18392    | -4.141  | 3.46e-05  | *** |
| v1          | 1.81390  | 0.13506    | 13.430  | < 2e-16   | *** |
| x1:y1       | -0.55228 | 0.30954    | -1.784  | 0.07439   | .   |
| x1:z1       | -0.18010 | 0.09969    | -1.807  | 0.07082   | .   |
| y1:z1       | -0.08484 | 0.16698    | -0.508  | 0.61138   |     |
| x1:v1       | -0.46747 | 0.18035    | -2.592  | 0.00954   | **  |
| y1:v1       | -0.42278 | 0.26239    | -1.611  | 0.10711   |     |
| z1:v1       | 3.30975  | 0.18462    | 17.928  | < 2e-16   | *** |
| x1:y1:z1    | 0.15699  | 0.32673    | 0.480   | 0.63088   |     |

The final model can be written as

$$\begin{aligned} \log(n) = & 3.93498 - 0.27621 \cdot x1 - 1.65646 \cdot y1 - 0.76156 \cdot z1 + 1.81390 \cdot v1 \\ & - 0.55228 \cdot x1y1 - 0.18010 \cdot x1z1 - 0.08484 \cdot y1z1 - 0.46747 \cdot x1v1 \\ & - 0.42278 \cdot y1v1 + 3.30975 \cdot z1v1 + 0.15699 \cdot x1y1z1 \end{aligned}$$

Comparing values in Table 7 with that in Table 5, we can easily find some equal values.

Table 8: Equality in LR and Log-linear model

| LR model  | Log-linear model | Estimates | Std. Error |
|-----------|------------------|-----------|------------|
| Intercept | v1               | 1.8139    | 0.1351     |
| X         | x1:v1            | -0.4675   | 0.1804     |
| Y         | y1:v1            | -0.4228   | 0.2624     |
| Z         | z1:v1            | 3.3098    | 0.1846     |

Thus, we can conclude that the two models give identical estimates and standard errors of the corresponding parameters.

## Index för kommentar

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2.1      These are not odds ratios.